Energy and AI 25 (2026) 100811

Contents lists available at ScienceDirect

Energy and AI

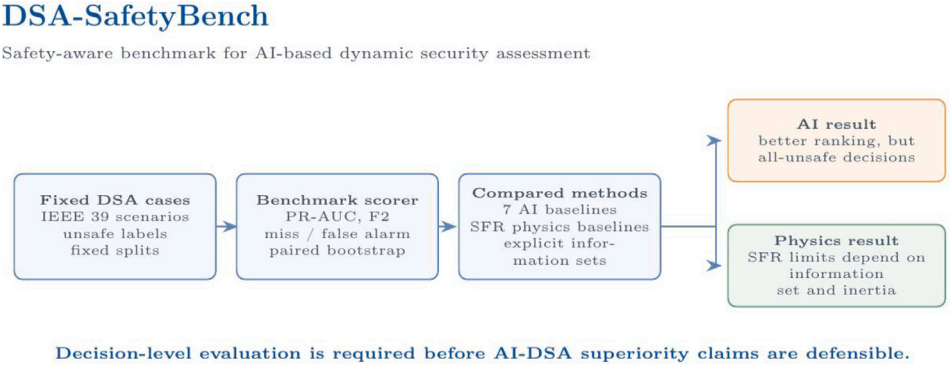
journal homepage: www.elsevier.com/locate/egyai

DSA-SafetyBench: A safety-aware benchmark for artificial-intelligence-based dynamic security assessment in energy systems

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GRAPHICAL ABSTRACT



HIGHLIGHTS

- DSA-SafetyBench scores AI-based DSA decisions, not only trajectories.
- Seven ML baselines produce all-unsafe calibrated decisions.
- Paired bootstrap finds no statistically separable architecture winner.
- SFR baselines expose information-set and effective-inertia limits.
- Artifacts include records, schema, scorer, and robustness scripts.

ARTICLE INFO

Keywords:
Dynamic security assessment
Artificial intelligence
Safety-aware benchmarking
Frequency security
Swing-frequency response
Dynamic state estimation

ABSTRACT

Benchmarks have accelerated progress in coding, mathematics, and language, but artificial-intelligence-based dynamic security assessment (DSA) lacks a safety-aware benchmark for operational decisions. We introduce DSA-SafetyBench, an IEEE 39-bus protocol that tests whether machine-learning models produce useful frequency-security decisions, not only accurate trajectories. It defines fixed splits, unsafe labels, validation-calibrated thresholds, decision-level metrics, repeated-seed paired-bootstrap uncertainty, per-scenario records, and swing-frequency-response (SFR) physics baselines with explicit information sets. On the base benchmark, regression-derived safety decisions are near-degenerate: models with acceptable trajectory errors rank unsafe events near the base rate and produce all-safe or all-unsafe policies. A three-class auxiliary safety head improves ranking, but five-seed evaluation does not separate architectures. On the expanded benchmark (820 scenarios, 219 unsafe), denser coverage raises PR-AUC yet all seven methods still collapse to all-unsafe hard decisions under F2-opt, target-recall, and fixed-alarm-budget tests. SFR baselines show complementary limits: ambient

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https://doi.org/10.1016/j.egyai.2026.100811
Received 21 May 2026; Received in revised form 31 May 2026; Accepted 9 June 2026
Available online 18 June 2026
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dynamic state estimation is weak, 5 s event-triggered fitting is delayed nowcasting, and nameplate inertia is not a stable anchor. Crucially, a leak-free event-triggered information set – each model sees only the first 0.5–2 s of the measured frequency response, aligned with the event-triggered physics estimator's measurement window – largely reduces the collapse and recovers non-trivial low-false-alarm operating points across IEEE 39, NPCC (140-bus), and WECC (179-bus, renewable-rich), with remaining threshold- and system-dependent limits. DSA-SafetyBench shows artificial-intelligence-based DSA claims require decision-level, multi-seed, physics-aware evaluation, and that event-triggered screening is a constructive path where pre-event descriptors are insufficient.

1. Introduction

Dynamic security assessment (DSA) is an operational screening problem: after a disturbance, an operator needs to know whether the power system will remain within security limits or whether preventive or corrective action is required. For frequency-security assessment, the critical outcomes are unsafe frequency nadirs – the lowest frequency reached after the event – and divergent trajectories. Average trajectory quality is secondary if the resulting decision is false-safe.

Within energy-system operation, this problem sits at the intersection of wide-area measurement, data-driven analytics, and physics-based decision support, making it a useful test case for evaluating artificial intelligence claims under operational security constraints.

Progress in other artificial intelligence domains is often measured through standardized benchmarks with fixed tasks, held-out splits, shared scoring scripts, and baseline leaderboards. Coding and mathematical-reasoning benchmarks do not merely report whether a model fits training data; they test whether model outputs solve the intended task under a defined scoring protocol. Artificial-intelligence-based DSA needs the same discipline. A DSA benchmark should ask not only whether a model predicts trajectories accurately, but whether it ranks unsafe cases, supports calibrated decisions, remains stable across seeds, and is compared against physics baselines under explicitly declared information sets.

Recent machine-learning-based DSA (ML-DSA) studies often report aggregate accuracy, trajectory mean-squared error, or nadir mean absolute error. These metrics are useful but incomplete under the class imbalance typical of simulated security datasets. A model can report high accuracy by predicting every case as safe, or low nadir error by fitting common stable trajectories while failing to rank rare unsafe cases. This paper treats that decision-level gap as the object of study.

The broader artificial-intelligence-evaluation literature has long warned that single-seed rankings, imbalanced labels, and threshold choices can change conclusions. The DSA-specific contribution is to show how those known risks manifest in power-system security, where asymmetric operational cost, trajectory-regression surrogates for security decisions, and physics baselines with limited applicability interact.

This paper introduces DSA-SafetyBench to make these validity checks explicit. The benchmark separates simulator information-set audits from decision-level scoring: historical or scenario features are credited only when they are causally connected to the simulated dynamics, and machine-learning forecasts are evaluated by unsafe-event ranking, calibrated hard decisions, and repeated-seed uncertainty rather than by trajectory error alone.

Physics baselines are treated with the same care. Aggregate swing-frequency response (SFR) methods require a scalar active-power disturbance and defensible dynamic parameters. Some scenarios, such as self-clearing bus faults, are not SFR-compatible. Others, such as load steps and generator-output-loss events, are SFR-compatible but expose dynamic-state-estimation (DSE) issues: multi-mode ambient dynamics, unstable nameplate-inertia anchoring, and early-window identifiability. The benchmark asks what each method can claim under the information it is actually allowed to use.

2. Contributions

1. We define DSA-SafetyBench, a safety-aware benchmark protocol and artifact package for ML-DSA. It includes fixed scenario splits, per-scenario prediction records, unsafe-event labels, validation-calibrated hard decisions, decision-level metrics, paired bootstrap uncertainty over repeated seeds, a prediction schema, a case inventory, and a reproducible benchmark scoring script.
2. We provide an IEEE 39-bus reference implementation with two benchmark streams: a base stream that reveals regression-derived decision weaknesses under sparse unsafe samples, and an expanded SFR-compatible stream with 820 load-step and generator-output-loss scenarios for matched physics comparisons.
3. We implement a baseline suite covering regression-only ML, auxiliary-safety-head ML, no-history and history-aware sequence models, and SFR-class physics baselines with explicit information-set accounting.
4. We show benchmark findings that are hidden by common ML-DSA reporting practice: regression-derived safety assessment can produce all-safe or all-unsafe policies, single-seed architecture rankings are not reliable, and denser unsafe coverage improves ranking without yielding statistically separable model winners.
5. We audit benchmark validity assumptions, including simulator history-channel causality and SFR disturbance compatibility. The audit shows that history channels are not causally consumed by the simulator, and the SFR-family results expose weak ambient DSE, near-oracle delayed nowcasting, competitive first-order envelope fitting, and strongly contingency-dependent effective inertia.
6. We identify a constructive path: a leak-free event-triggered information set, in which each model observes only the first 0.5–2 s of the measured frequency response. This measurement window is aligned with the event-triggered physics estimator; it largely reduces the all-unsafe collapse and recovers usable low-false-alarm operating points. This result holds across IEEE 39, NPCC (140-bus), and WECC (179-bus, renewable-rich), which also externally validate the protocol under a common early-response information set.

3. Related work

3.1. Artificial intelligence and power-system benchmarks

Power-system research already has several influential benchmark traditions, but they target different operational questions than ML-DSA safety decisions. PGLib-OPF standardizes AC optimal-power-flow formulations and network data so optimization algorithms can be compared on common cases [1]. L2RPN and Grid2Op turn transmission-grid operation into an interactive topology-control benchmark for reinforcement-learning agents under operational constraints [2]. PowerGym provides an OpenAI-Gym-style benchmark for distribution-system volt-var control, with IEEE feeder variants and physical voltage and loss objectives [3]. Public transient-stability datasets for the IEEE New

England 39-bus system provide useful data for ML-based TSA classification and feature studies [4], and recent DSA dataset-generation work emphasizes sampling near the security boundary to improve data-driven assessment [5].

These efforts establish the value of shared cases, public datasets, and standardized evaluation in power systems. They do not, however, provide a benchmark protocol for safety-aware ML-DSA decisions with fixed scenario identifiers, repeated training seeds, per-scenario prediction records, hard-decision calibration, unsafe-miss and false-alarm reporting, paired scenario-level uncertainty, and physics-baseline information-set accounting. DSA-SafetyBench is designed to fill that gap rather than replace OPF, topology-control, volt-var, or general TSA datasets.

3.2. Machine-learning dynamic security assessment

ML-based DSA has been studied with feedforward, convolutional, recurrent, graph-based, and attention-based architectures that map pre-contingency operating conditions or short post-contingency measurements to stability labels, trajectory forecasts, or security margins [6–10]. These approaches are attractive because they can amortize expensive time-domain simulation into fast online inference. However, many studies emphasize aggregate accuracy or regression quality, which can obscure unsafe-event performance when unsafe samples are sparse.

Trajectory-prediction formulations are especially relevant here because nadir and RoCoF can be derived from predicted frequency trajectories. The results show that this derivation is not automatically reliable: a trajectory predictor can have acceptable average error while failing to rank rare unsafe events. The benchmark therefore separates trajectory fidelity from safety ranking.

The benchmark is not intended to exhaust the full design space of rare-event learning. Cost-sensitive losses, ranking losses, conformal prediction, selective classification, and uncertainty-aware screening may improve operational decision quality. The goal here is narrower: to test common trajectory-regression and auxiliary-safety-head formulations under safety-aware metrics and explicitly declared information sets.

3.3. Sequence models and causal grid histories

Long-horizon sequence models, including LSTM [11], Transformer [12], TCN [13], neural ODE or liquid-time-constant models [14–16], and state-space models such as Mamba [17,18], offer natural tools for history-aware DSA. The benchmark keeps these architectures in the comparison but does not claim a causal-history advantage where the simulator data do not support one. Long-context modeling is meaningful only if historical channels actually drive the simulated or measured dynamics.

3.4. SFR, dynamic state estimation, and information sets

Aggregate SFR models approximate system frequency response with low-order swing and governor dynamics [19,20]. They are interpretable and operationally attractive, but they require a well-defined active-power disturbance and effective dynamic parameters. DSE can estimate such parameters from ambient or early post-event measurements, but the information set matters [21,22]. Fitting a model from the full post-event trajectory gives an oracle upper bound, while ambient and event-triggered estimates face model-class mismatch, measurement noise, window-length limits, and parameter identifiability.

The benchmark therefore distinguishes SFR-oracle, ambient SFR-DSE, event-triggered SFR-DSE, noisy event-triggered SFR-DSE, and misspecified SFR-DSE. This avoids the common practice of comparing ML against a single "SFR" row whose assumptions are not specified.

3.5. Evaluation under rare unsafe events

Operational DSA is a rare-event screening task. Accuracy is dominated by the safe class, and thresholded metrics depend strongly on calibration. Precision–recall analysis, F-score variants emphasizing recall, unsafe miss rate, false alarm rate, repeated seeds, and paired uncertainty estimates are therefore central to the protocol [23–25]. Selective classification and conformal prediction provide related safety-critical ML tools for abstention and uncertainty quantification, but DSA-SafetyBench focuses on the more direct operating question of whether a model can rank and threshold unsafe events under fixed alarm budgets [26,27]. The five-seed results show that single-seed architecture rankings can invert on the same benchmark, which is itself a methodological warning for ML-DSA comparison studies.

4. DSA-SafetyBench design

4.1. Benchmark task and scoring protocol

DSA-SafetyBench defines a model-facing task, a scoring protocol, and a per-scenario record format. A model receives the scenario inputs allowed by its information set and returns an unsafe score, an optional calibrated hard decision, and any regression output such as predicted nadir. The benchmark scorer evaluates those submissions against fixed scenario identifiers, fixed splits, and hidden or released labels depending on the evaluation setting. The full per-scenario record includes scenario identifier, split, unsafe score, calibrated hard decision, decision source, predicted nadir, true nadir, unsafe label, unsafe type, active-power disturbance, model name, and training seed. The required fields for the benchmark scorer are scenario identifier, unsafe score, and unsafe label; the remaining fields enable calibrated-decision metrics, nadir MAE, class-stratified reporting, and SFR information-set audits. Fig. 1 summarizes the workflow from time-domain simulation through declared information sets, validation thresholding, decision metrics, paired-bootstrap uncertainty, and released per-scenario records.

The reference task is online frequency-security assessment after a contingency. For each scenario, the target trajectory is the center-of-inertia frequency deviation over the post-event simulation window. A scenario is labeled unsafe if the trajectory crosses the frequency-security nadir threshold (-0.1 Hz) or diverges; otherwise it is labeled safe. The -0.1 Hz value is a benchmark security threshold chosen to create a controlled rare-event screening task with measurable class separation; it is not intended as a universal grid-code, protection, or under-frequency-load-shedding criterion. Unsafe cases are further stratified into nadir-threshold-violating and divergent outcomes. This three-class label is used for safety supervision and for stratified reporting, while aggregate binary unsafe labels are used for operational metrics such as unsafe miss and false alarm rates.

The operational error asymmetry is central: a false-safe decision can expose the system to an unsafe event, while a false-unsafe decision is conservative but increases alarm burden. We therefore do not use aggregate accuracy as a primary metric. Instead, we report PR-AUC for unsafe-event ranking, F2-opt to emphasize unsafe recall, validation-calibrated hard decisions, unsafe miss rate, false alarm rate, nadir MAE, and per-class recall for nadir-threshold-violating and divergent cases. F2 is the F-score with $\beta = 2$, weighting recall more heavily than precision.

4.2. ML models and safety supervision

We evaluate standard sequence and snapshot ML baselines under a matched train/validation/test split: LiquidMamba variants, LiquidPure without history, LSTM, Transformer+LSTM, Mamba+LSTM, TCN, and CNN-1D. The regression-only form of each model predicts a frequency trajectory and derives a safety score from the predicted nadir.

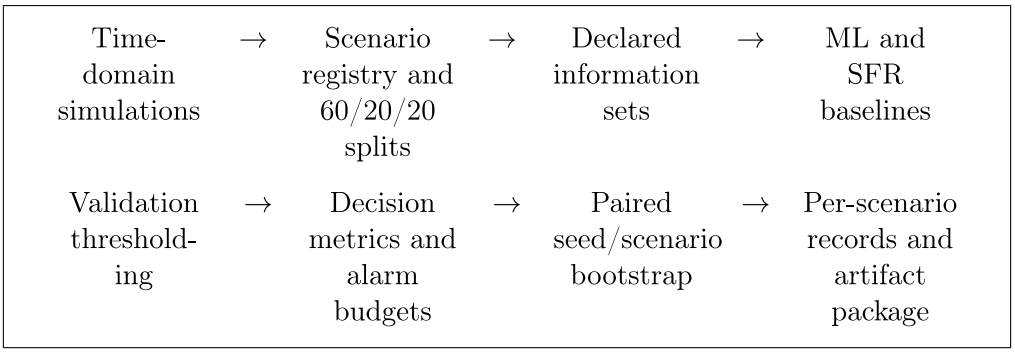


Fig. 1. DSA-SafetyBench workflow. The protocol starts from time-domain simulations, fixes scenario identities and train/validation/test splits, declares the information available to each model or physics baseline, calibrates hard decisions on validation data, reports decision-level safety metrics, estimates uncertainty with paired repeated-seed bootstrapping, and releases per-scenario records for reuse.

The safety-aware form keeps the trajectory head but adds a parallel three-class auxiliary safety head.

The auxiliary head operates on the event-decoder hidden sequence, not on the predicted nadir. For each model, the hidden sequence is pooled by concatenating its temporal mean, temporal maximum, first hidden state, and final hidden state. The pooled vector is passed through LayerNorm, a linear projection from 4d to d, ReLU, and a final linear layer with three logits for the safe, nadir-threshold-violating, and divergent classes. The protocol does not feed predicted nadir into the safety head, because ablation experiments showed that predicted nadir is least reliable on the unsafe tail and can anchor the classifier to regression errors.

The training objective combines trajectory loss and auxiliary safety focal loss. The focal-loss defaults are global alpha 0.5 and gamma 2.0, with class labels drawn from the three-way safety target. Checkpoint selection uses validation safety PR-AUC rather than trajectory MSE when safety supervision is active. Hard decisions are calibrated on the validation split by selecting the unsafe-score threshold that maximizes F2. This design intentionally separates two questions that are often conflated: whether a model predicts average trajectories well, and whether it ranks rare unsafe events well enough for operational screening.

Auxiliary-head architecture and focal-loss settings are held fixed across models to avoid uneven hyperparameter search. The results therefore test whether simple, explicit safety supervision is sufficient to turn common ML-DSA architectures into useful operational screeners; they are not a final statement about the best possible rare-event classifier.

In addition to the regression-only and auxiliary-head forms, we evaluate an event-triggered form under a deliberately restricted, leak-free information set. The model observes only the first τ seconds of the measured center-of-inertia frequency response, supplied through a derived observation channel and an explicit observation mask, with all later samples masked and with no fault location, disturbance magnitude, generator identity, or inertia value provided. Because the frequency nadir forms several seconds after the disturbance, the first τ seconds do not include the final nadir or divergence outcome, so the model must infer severity from the early rate of change and slope; the information set is therefore leak-free. This measurement window is aligned with the event-triggered SFR-DSE physics baseline defined below. SFR-DSE additionally requires a characterized active-power disturbance, so the comparison is matched on early-response timing while retaining explicit input differences. We sweep $\tau \in \{0.5, 1.0, 2.0\}$ s, mirroring the SFR-DSE event-triggered window family; results are reported in the event-triggered screening subsection of the Results.

4.3. Repeated seeds and bootstrap protocol

Single-seed comparisons are fragile when the unsafe set is sparse. We therefore repeat the top-method benchmark over five random seeds and save per-scenario records covering identity (seed, scenario identifier, model name, and split), prediction (unsafe score, calibrated hard decision, decision source, and regression nadir), ground truth (unsafe label, unsafe type, and true nadir), and disturbance characterization (active-power imbalance). The same scenario identities are evaluated across seeds; the repeated-seed dimension captures training stochasticity rather than a different test set. The two resampling dimensions correspond to two distinct uncertainty sources: the repeated training seeds isolate epistemic uncertainty arising from model initialization and optimization, while resampling scenarios within each seed reflects the aleatory variability of the disturbance population. This distinction is consistent with sociotechnical risk-analysis work, where expert judgment, model assumptions, data limitations, and scenario variability are treated as separate contributors to uncertainty [28,29]. Reporting both, rather than a single-seed point estimate, is precisely what exposes the non-replicable architecture rankings that single-seed studies miss.

For paired comparisons, we use the saved per-scenario records so that method differences are computed on matched scenario identities. The protocol is a two-level paired bootstrap: resample scenarios within each seed, compute method-pair differences per seed, then average seed-level differences and bootstrap across seeds. Confidence intervals are percentile intervals computed with 2000 bootstrap replicates. For the base-benchmark auxiliary-head summary, where only seed-level aggregate intervals were available, Table 4 reports seed-resampling intervals and treats them as robustness diagnostics; the expanded-benchmark pairwise intervals use the saved per-scenario records. Because all top-method confidence intervals include zero on the base benchmark, no architecture winner is claimed from that dataset. Multiple-comparison correction is therefore not decision-changing: all unadjusted pairwise intervals already fail to separate the top methods.

This no-winner conclusion is interpreted with the available unsafe-event support in view. With approximately 10 unsafe events per stratified test split in the base benchmark, single-seed and even five-seed paired confidence intervals are inherently wide. The expanded benchmark enriches unsafe coverage to 26.7% (219 of 820 scenarios; 124/420 load-step and 95/400 generator-output-loss) to test whether the no-winner finding persists under tighter intervals or whether some methods separate when the unsafe set is denser.

4.4. Simulator and information-set audit

A common motivation for history-aware DSA is the hypothesis that 24-hour SCADA histories encode path-dependent latent stress unavailable to snapshot models. We audit that assumption before interpreting any model advantage.

The data generator contains history-like fields such as load history, latent stress, and AGC/governor proxies. A controlled audit varies

Table 1
Controlled audit of candidate history and stress mechanisms in the IEEE 39-bus data generator. A mechanism is marked observable only if changing its control value produces measurable trajectory variation in the transient simulation. The audit shows that only inertia scaling is causally consumed by the simulator; the history-like channels are ML inputs but do not drive ANDES transient dynamics.

Mechanism	Status	Levels	Nadir spread (Hz)	SFR α spread	Interpretation
Damping scale	No observable effect	3	0.000	0.000	Control value changes, but the trajectory and fitted response do not measurably change.
Inertia scale	Observable	5	0.122	1.111	Control value changes aggregate inertia and produces measurable nadir and fitted-response variation.
Latent stress	No observable effect	3	0.000	0.000	Control value changes, but the trajectory and fitted response do not measurably change.
History load profile	Not consumed	–	–	–	Written as an ML history feature, but not applied to ANDES PQ injections before transient simulation.
AGC/governor history proxy	Not consumed	–	–	–	Written as synthetic history inputs, but not consumed by simulator governor or AGC dynamics.

each candidate stress mechanism and measures the resulting trajectory variation. Table 1 summarizes the result: inertia scaling produces measurable nadir and fitted-response variation; damping scaling and latent stress have no observable effect; and the load-history and AGC/governor-history proxy fields are written as ML inputs but are not consumed by ANDES transient dynamics. These fields may be correlated with scenario metadata, but they do not drive the post-event dynamics. Therefore, any performance difference between history-aware and no-history models on the base benchmark cannot be interpreted as evidence of learned physical path dependence.

This audit determines the information-set rules used in the benchmark. ML methods may use scenario features and trajectories available in the dataset, but no claim is made that history encodes causal latent state unless the simulator actually consumes that history. Physics baselines are evaluated only when their required disturbance information is available without using the post-event response.

4.5. SFR-class physics baselines

Self-clearing bus-fault scenarios are not SFR-compatible by construction. A second-order aggregate SFR model requires a scalar active-power disturbance, ΔP_d, and follows the standard aggregate frequency-response abstractions used for nadir and primary-frequency-response analysis [19,20]. A temporary bus fault produces a spatial and time-varying voltage depression whose net frequency impact depends on network and protection dynamics. We therefore block SFR rows on the base benchmark rather than infer ΔP_d from the response.

The expanded benchmark introduces SFR-compatible load-step and generator-output-loss scenarios. For load steps, ΔP_d is the imposed active-load increase. For generator-output-loss scenarios, ΔP_d is the specified fraction of the pre-event generator active power, implemented as a partial active-power loss rather than necessarily as a full generator disconnection. Partial generator-output loss represents runback, derating, mechanical-power reduction, or other partial active-power-loss events; it also provides a controllable disturbance magnitude for spanning safe and unsafe regimes. A calibration study found that 25%–100% generator-output loss was uniformly too severe for the tested generators, so the balanced dataset uses 15, 30, 45, and 60%.

Inertia scaling over 0.7–1.3 is interpreted as a controlled proxy for changes in online synchronous generation, retirements, commitment, or synchronous-condenser availability rather than as a literal single operating action. The final expanded dataset contains 420 load-step scenarios and 400 generator-output-loss scenarios.

The SFR family is separated by information set. SFR-oracle fits a damped second-order SFR response to the full post-event trajectory and serves only as an upper-bound anchor; it is not deployable because it uses the response being evaluated. Ambient SFR-DSE estimates shape parameters from pre-event stochastic-load frequency measurements, reflecting the broader DSE objective of tracking dynamic states and parameters from time-synchronized measurements [21,22]. Event-triggered SFR-DSE fits parameters from the first 1, 2, or 5 s of measured post-disturbance response. Noisy variants inject deterministic noise at three times the nominal measurement-noise level into the event window, and misspecified variants fit a first-order monotone step model to the same windows.

This separation makes the operational assumptions visible rather than hiding them in a single "SFR" row. Table 2 summarizes the per-method allowed inputs, decision timing, and deployability so that the machine-learning and physics comparisons are read under explicit information-set declarations.

4.6. External systems for generalization

To test whether DSA-SafetyBench instantiates beyond the IEEE 39-bus reference, we apply the same generation, labeling, splitting, and scoring pipeline to two larger systems: the 140-bus NPCC system and the 179-bus WECC system, the latter a renewable-rich Western Interconnection case. Both are screened before use with a nadir-spread criterion: the disturbance family must produce at least a 0.05 Hz spread in frequency nadir across the operating range, so that the benchmark contains a non-degenerate mix of safe and unsafe outcomes. Under the generator-trip and load-step disturbance classes both clear this screen (NPCC 0.106 Hz, WECC 0.058 Hz); self-clearing bus faults produce negligible nadir spread and are unsuitable for this screen.

The disturbance class is adapted per system because the generator-output-loss generator requires a TGOV1N governor model that NPCC (TGOV1) and WECC (IEEEG1) do not provide. NPCC therefore uses generator-trip scenarios and WECC uses load-step scenarios at convergence-screened buses, both governor-agnostic and both validated to produce a usable safe/unsafe mix (NPCC 17% unsafe, WECC 31% unsafe in the test split). These streams are scoped: they use short ambient windows and are intended for the leak-free event-triggered ML comparison reported in the Results, not for a full ambient SFR-DSE physics replication. They serve as external validation of the protocol under a common early-response information set rather than as replacements for the IEEE 39-bus reference benchmark, which remains the primary result.

5. Results

5.1. Regression-derived safety is degenerate on the base benchmark

On the base IEEE 39-bus dataset, the seed-42 stratified test split contains 83 scenarios, including 10 unsafe cases. A ground-truth nadir score achieves PR-AUC 0.722 and F2-opt 0.833, confirming that the safety labels are learnable from a sufficiently accurate nadir signal. The learned regression-derived safety scores do not preserve that signal. Across the regression-only model forms, PR-AUC clusters near the unsafe base rate (10/83 = 0.120), spanning 0.092 for LiquidMamba to

Table 2
Information-set and decision-timing rules for DSA-SafetyBench baselines. The table separates screening methods, non-deployable upper bounds, pre-event estimators, and delayed event-triggered nowcasts so that ML and physics baselines are not compared under hidden information asymmetries.

Method	Allowed inputs	Uses measured post-event response?	Decision timing	Deployability
Regression ML	Scenario, history, and fault descriptor features allowed by the benchmark split; predicted trajectory is model output.	No	Screening once scenario inputs are available.	Deployable screening
Auxiliary-head ML	Same scenario, history, and fault descriptors; unsafe score is produced from the decoder hidden representation.	No	Screening once scenario inputs are available.	Deployable screening
SFR-oracle	Full observed post-event frequency response and active-power disturbance.	Yes, full response	After the event record.	Non-deployable upper bound
Ambient SFR-DSE	Pre-event ambient frequency measurements and active-power disturbance.	No	Parameters before the event; decision once disturbance is known.	Deployable if disturbance is characterized
Event-triggered SFR-DSE	First 1, 2, or 5 s of measured post-disturbance response and active-power disturbance.	Yes, early response	After selected measurement window.	Delayed nowcast
Noisy/misspecified SFR-DSE	Same event-triggered window with injected measurement noise or reduced-order model.	Yes, early response	After selected measurement window.	Robustness and model-class baseline

Table 3
Regression-derived safety decisions on the IEEE 39-bus base-benchmark test split. The test set contains 83 scenarios, including 10 unsafe events. PR-AUC is near the unsafe base rate for all learned models, and thresholded decisions produce either all-safe behavior or excessive false alarms. Unsafe miss is the false-safe rate.

Model	PR-AUC	F2-opt	Unsafe miss	False alarm	Nadir MAE (Hz)
LiquidMamba (tau-gate)	0.092	0.348	1.000	0.000	0.129
LiquidPure (No History)	0.151	0.357	1.000	0.000	0.192
LSTM (History)	0.112	0.420	1.000	0.000	0.021
Mamba+LSTM	0.109	0.439	1.000	0.000	0.029
Transformer+LSTM	0.136	0.459	0.000	1.000	0.083
TCN (History)	0.101	0.417	1.000	0.041	0.157
CNN-1D (Snapshot)	0.148	0.495	1.000	0.000	0.175

0.151 for LiquidPure; Table 3 reports the full PR-AUC, F2-opt, unsafe miss, false-alarm, and nadir-MAE results.

Most regression-derived models predict no unsafe cases at the benchmark's −0.1 Hz threshold, yielding 100% unsafe miss rates. Transformer+LSTM moves in the opposite direction, producing a 100% false-alarm rate. Nadir MAE does not resolve the problem: LSTM and Mamba+LSTM have low nadir MAE but still near-random unsafe-event ranking. The primary issue is therefore not trajectory error alone; it is the mismatch between regression training and rare unsafe-event decision quality.

5.2. Decision patterns under regression-derived safety

We dedicate a subsection to decision patterns because they explain why aggregate trajectory metrics were misleading. Under regression-derived safety, the models do not produce a single uniform error pattern. Several architectures, including LSTM and Mamba+LSTM, learn stable-looking trajectory regressors whose thresholded nadirs produce all-safe decisions. This produces low alarm burden but catastrophic unsafe miss rates. Transformer+LSTM shows the opposite pattern on the seed-42 split: it marks every case unsafe, eliminating unsafe misses at the cost of a 100% false-alarm rate.

This decision-level mode collapse is more operationally relevant than whether the average trajectory appears smooth. These results show that recurrent, attention, and hybrid sequence models occupy different regions of the safety-accuracy tradeoff: some preserve trajectory accuracy while suppressing rare unsafe excursions, while others become overly sensitive and alarm on nearly all scenarios. The safety-aware metrics sharpen that observation. Mode collapse is not merely a visual artifact in predicted trajectories; it becomes a degenerate operating policy when a regression model is converted into a binary security decision.

The mechanism is consistent with gradient starvation on the unsafe tail: trajectory-regression loss is dominated by stable trajectories, so the optimizer reduces average error while allocating little capacity to rare nadir excursions or divergent cases. When a hard safety decision is then derived from the predicted nadir, small systematic errors near the security threshold become one-sided decision failures: either all-safe behavior when unsafe excursions are smoothed away, or all-unsafe behavior when the learned trajectory bias shifts nadirs below the threshold.

The auxiliary safety head reduces but does not eliminate this architecture-level fragility. It gives each model a direct gradient on unsafe outcomes, but the five-seed benchmark still shows unstable rankings and overlapping paired confidence intervals. The decision-level conclusion is therefore broader than any one architecture: regression-only training under sparse unsafe samples is not a reliable route to operational DSA screening.

5.3. Auxiliary safety supervision helps but does not establish a winner

Adding the auxiliary three-class safety head improves unsafe-event ranking for several models. In the seed-42 single-seed analysis on the base benchmark, Transformer+LSTM and LiquidPure reach PR-AUC around 0.59, while LiquidMamba reaches 0.401. Calibration also exposes different operating points: Transformer+LSTM can reach 20% unsafe miss with 16.4% false alarm, while LiquidMamba reaches 30% unsafe miss but with 49.3% false alarm. The base-benchmark five-seed evaluation reported in Table 4 uses the four highest-PR-AUC methods from this single-seed pass, while the expanded-benchmark evaluation below uses the full seven-model baseline suite over five seeds to remove the seed-42 preselection confound entirely.

The single-seed ranking is not reliable. In the base-benchmark five-seed evaluation of the top four methods, LiquidMamba has the highest mean PR-AUC at 0.408, but paired differences versus the other top methods are not statistically separated. Mean PR-AUC values are 0.408 for LiquidMamba, 0.370 for LSTM, 0.361 for Transformer+LSTM, and

Table 4
Base-benchmark five-seed auxiliary safety-head summary for the four methods selected from the seed-42 screening pass. Bootstrap intervals resample seeds and should be read as robustness diagnostics rather than population guarantees; all paired PR-AUC and F2-opt differences among the top four include zero.

Model	PR-AUC	PR-AUC 95% CI	F2-opt	F2 95% CI	Unsafe miss	False alarm
LiquidMamba (tau-gate)	0.408	[0.257, 0.534]	0.532	[0.456, 0.608]	0.220	0.512
LSTM (History)	0.370	[0.233, 0.506]	0.530	[0.459, 0.600]	0.300	0.422
Transformer+LSTM	0.361	[0.287, 0.476]	0.547	[0.487, 0.612]	0.440	0.241
LiquidPure (No History)	0.280	[0.122, 0.455]	0.489	[0.418, 0.576]	0.220	0.614

Table 5
Expanded-benchmark five-seed ML safety evaluation across all seven baselines (no seed-42 preselection). Panel A reports model-level means. Panel B reports representative paired scenario-within-seed bootstrap PR-AUC differences using 2000 percentile bootstrap replicates; all 21 pairwise PR-AUC intervals include zero, and F2-opt differences are zero by construction because every method selects the all-unsafe hard decision (F2 = 0.647 = unsafe-base-rate F2). Zero-difference rows are retained to show tied saved unsafe-score arrays for those model pairs. Panel C reports a test-set operating-point diagnostic: even when thresholds are selected to maximize unsafe recall subject to false-alarm-rate budgets of 5%, 10%, and 20%, every method achieves zero unsafe recall.

Panel A: Model-level means (seven models, five seeds each)

Model	PR-AUC	F2-opt	Unsafe miss	False alarm	Nadir MAE (Hz)
Mamba+LSTM	0.686	0.647	0.000	1.000	0.123
LSTM (History)	0.653	0.647	0.000	1.000	0.124
LiquidPure (No History)	0.653	0.647	0.000	1.000	0.258
CNN-1D (Snapshot)	0.630	0.647	0.000	1.000	0.211
TCN (History)	0.630	0.647	0.000	1.000	1.091
Transformer+LSTM	0.590	0.647	0.000	1.000	0.101
LiquidMamba (tau-gate)	0.568	0.647	0.000	1.000	0.289

Panel B: Representative paired bootstrap PR-AUC differences

Comparison	Δ PR-AUC	95% CI	Interpretation
Mamba + LSTM − LiquidMamba	+0.032	[−0.010, +0.082]	Not statistically separated
Mamba + LSTM − LSTM	+0.006	[−0.013, +0.036]	Not statistically separated
Mamba + LSTM − LiquidPure	+0.006	[−0.013, +0.036]	Not statistically separated
Mamba + LSTM − Transformer+LSTM	+0.036	[+0.000, +0.092]	Borderline; CI touches zero
Mamba + LSTM − CNN-1D	+0.016	[−0.001, +0.063]	Not statistically separated
LSTM − LiquidPure	+0.000	[+0.000, +0.000]	Identical scores in this run
LSTM − LiquidMamba	+0.025	[−0.006, +0.076]	Not statistically separated
LiquidPure − LiquidMamba	+0.025	[−0.006, +0.076]	Not statistically separated
CNN-1D − TCN	+0.000	[+0.000, +0.000]	Identical scores in this run

Panel C: Fixed false-alarm-budget diagnostic

False-alarm budget	Best mean unsafe recall	Mean unsafe miss	Models
5%	0.000	1.000	All seven methods
10%	0.000	1.000	All seven methods
20%	0.000	1.000	All seven methods

0.280 for LiquidPure. All pairwise PR-AUC and F2-opt confidence intervals include zero. Thresholded metrics show tradeoffs rather than dominance: LiquidPure and LiquidMamba have lower unsafe miss rates but high false-alarm rates, while Transformer+LSTM has lower false alarms but higher unsafe miss. Table 4 details the base-benchmark five-seed top-four summary.

This is the principal ML finding for the base benchmark. Safety supervision changes the apparent model landscape, but the data do not support a robust architecture winner. The architecture ranked first depends on which seed is reported, which directly challenges single-seed ML-DSA comparison claims.

5.4. The expanded benchmark tests whether denser unsafe coverage stabilizes ML rankings

On the expanded dataset (820 scenarios spanning load-step and generator-output-loss classes), all seven ML baselines were trained over five seeds without seed-42 preselection, which avoids the selection-bias confound of restricting the multi-seed comparison to top-4 candidates chosen on a single seed. The five-seed mean PR-AUC values span Mamba+LSTM at 0.686, LSTM at 0.653, LiquidPure at 0.653, CNN-1D at 0.630, TCN at 0.630, Transformer+LSTM at 0.590, and LiquidMamba (tau-gate) at 0.568. F2-opt is identical at 0.647 across all seven methods because every model selects the trivial all-unsafe operating point at the validation-calibrated threshold. All 21 pairwise PR-AUC bootstrap intervals include zero; even the largest separation, Mamba+LSTM minus LiquidMamba at +0.032, has 95% CI [−0.010, +0.082]. Mamba+LSTM has the highest mean PR-AUC, but it was not among the initial top-4 selected from seed 42, which directly illustrates the seed-42 preselection bias that the full seven-model run removes. Table 5 reports the complete model list, representative pairwise CIs, and the fixed-false-alarm diagnostic, and Figs. 2 and 3 show the per-seed PR curves and rank heatmap.

LiquidMamba has the lowest mean PR-AUC among the seven baselines, but the paired CIs versus every other model include zero. This is evidence against a LiquidMamba-centered claim, not evidence for a statistically proven LiquidMamba disadvantage. Conversely, Mamba+LSTM has the highest mean PR-AUC at 0.686, but its CIs versus LSTM, LiquidPure, CNN-1D, and TCN also include zero, so no statistically separable leading architecture is identified.

The more consequential expanded-benchmark result is decision-threshold behavior. Despite mean PR-AUCs ranging from 0.568 to 0.686 across the seven models, every model achieves an F2-opt of exactly 0.647, the F2 of the trivial all-unsafe policy under the test unsafe base rate of 44/164 = 0.268. Every model therefore has unsafe miss rate 0.0 and false alarm rate 1.0 under the validation-calibrated F2-opt hard decision. A secondary test-set-calibrated target-recall analysis, which eliminates validation-set mismatch by selecting thresholds directly on

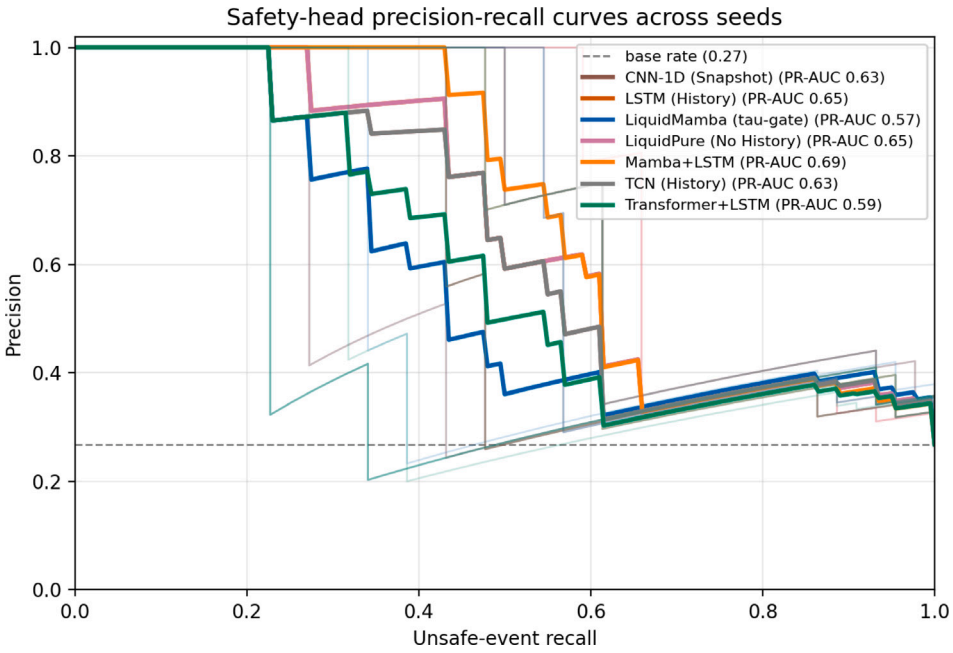


Fig. 2. Precision–recall curves for all seven auxiliary safety-head models on the expanded benchmark. Faint lines show individual training seeds and thick lines show seed-averaged behavior. The dashed horizontal line is the unsafe-event base rate. Denser unsafe coverage improves ranking relative to the base benchmark, but the methods remain statistically inseparable and hard-decision calibration selects all-unsafe policies.

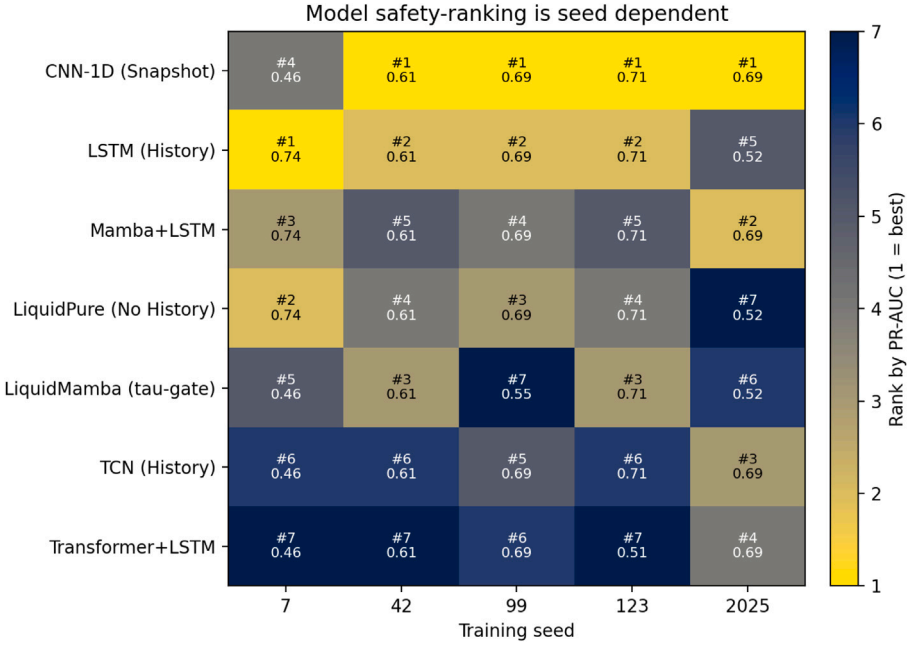


Fig. 3. Seed-level ranking heatmap for all seven auxiliary safety-head models on the expanded benchmark. Cells report rank by PR-AUC for each training seed. The ranking remains seed-sensitive, and paired bootstrap intervals do not support a statistically separable architecture winner.

the test scores, also selects all-unsafe decisions for every method and seed at 0.90 and 0.95 target recall.

An alarm-budget analysis reaches the same conclusion from the opposite direction. When thresholds are selected directly on the test scores to maximize unsafe recall subject to false-alarm-rate budgets of 5%, 10%, or 20%, every ML method has mean unsafe recall 0.0 across the five seeds. In these score distributions, the first threshold that recovers unsafe cases also alarms on roughly half of the safe cases. Monotone probability calibration, such as Platt scaling or isotonic calibration, can change probability values but cannot create a low-false-alarm operating point when the score ranking itself has this geometry. The PR-AUC improvement therefore does not translate into an operationally useful operating point under F2-opt, target-recall, or fixed-alarm-budget calibration.

5.5. Threshold sensitivity

The all-unsafe operating point could in principle be an artifact of the specific −0.1 Hz benchmark threshold. To test this, we re-derive unsafe labels at −0.05 Hz, −0.10 Hz, and −0.20 Hz directly from the

Table 6
Threshold sensitivity of the expanded-benchmark ML safety evaluation. For each candidate threshold, unsafe labels are re-derived from the saved per-scenario true-nadir values, and metrics are recomputed on the existing seven-model five-seed score arrays without retraining. Last column counts the number of models for which a majority of seeds still maximize F2 by predicting all-unsafe.

Threshold (Hz)	Mean unsafe base rate	Mean F2-opt (across models)	Models selecting all-unsafe
−0.05	0.635	0.897	7/7
−0.10	0.268	0.647	7/7
−0.20	0.030	0.211	1/7

Table 7
Calibration-policy comparison on the expanded benchmark. Each cell reports mean unsafe recall (left) and mean false-alarm rate (right) across five seeds, for one (model, policy) pair. F2-opt, target-recall (TR), and false-alarm-budget (FAR) policies are all derived directly from the saved test scores without retraining. Across all seven models, every recall-prioritizing policy selects all-unsafe decisions (recall = 1.0, FAR = 1.0), and every false-alarm-constrained policy selects zero-recall decisions.

Model	F2-opt	TR90	TR95	FAR 5%	FAR 10%	FAR 20%
Mamba+LSTM	1.000/1.000	1.000/1.000	1.000/1.000	0.000/0.000	0.000/0.000	0.000/0.000
LSTM (History)	1.000/1.000	1.000/1.000	1.000/1.000	0.000/0.000	0.000/0.000	0.000/0.000
LiquidPure (No History)	1.000/1.000	1.000/1.000	1.000/1.000	0.000/0.000	0.000/0.000	0.000/0.000
CNN-1D (Snapshot)	1.000/1.000	1.000/1.000	1.000/1.000	0.000/0.000	0.000/0.000	0.000/0.000
TCN (History)	1.000/1.000	1.000/1.000	1.000/1.000	0.000/0.000	0.000/0.000	0.000/0.000
Transformer+LSTM	1.000/1.000	1.000/1.000	1.000/1.000	0.000/0.000	0.000/0.000	0.000/0.000
LiquidMamba (tau-gate)	1.000/1.000	1.000/1.000	1.000/1.000	0.000/0.000	0.000/0.000	0.000/0.000

saved per-scenario true-nadir records, recompute PR-AUC and the test-set-calibrated F2-opt operating point, and check whether the all-unsafe result persists at each threshold. No retraining is required.

Table 6 reports the per-model summary at each threshold. At −0.05 Hz and −0.10 Hz, all seven models select the trivial all-unsafe operating point on a majority of seeds. At the much more selective −0.20 Hz threshold, the unsafe base rate drops to a small minority and only one of the seven models still maximizes F2 by predicting all-unsafe; the others avoid that specific operating point but still have low absolute PR-AUC. The threshold-collapse finding is therefore robust across the operationally realistic threshold range and is not an artifact of the chosen −0.1 Hz label.

5.6. Calibration-policy robustness

The operational question is whether any post-hoc calibration policy can recover a useful operating point from these score distributions. We compute the operating point of every (model, seed) under six policies: F2-opt, target recall 0.90, target recall 0.95, and false-alarm budgets 5%, 10%, and 20%. The first three policies trade false alarms for recall; the last three constrain false alarms and report whatever recall remains. Together they bound the achievable operating-point space under monotone score transformations.

Table 7 reports mean unsafe recall and mean false-alarm rate by (model, policy). Every model and every policy falls into one of two undesirable operating regimes: (i) under F2-opt, target-recall 0.90, and target-recall 0.95, every method achieves unsafe recall 1.0 at the cost of false-alarm rate 1.0, the trivial all-unsafe operating point; (ii) under false-alarm budgets of 5%, 10%, and 20%, every method achieves zero unsafe recall, because the first threshold that begins to identify unsafe cases also alarms on a non-trivial fraction of safe cases. Since these policies span the operationally useful boundary of monotone calibration, and Platt and isotonic calibration both preserve the score ordering, no monotone calibration can repair the operating point on these scores.

5.7. History channels need causal validation

The data do not support the history-aware claim. LiquidPure, which has no history input, can match or exceed LiquidMamba in single-seed safety ranking, and the five-seed benchmark does not statistically separate LiquidMamba from the no-history baseline. More importantly, the simulator audit shows that the history channels are not causal inputs to the dynamic simulation. Thus, even a statistical advantage would not prove path-dependent latent-state learning unless the simulator history pathways were made causal and the result replicated.

We therefore do not claim that 24-hour history captures latent stress in the base benchmark. History-aware modeling remains a possible future direction; it is not a demonstrated contribution of this dataset.

5.8. SFR applicability is scenario-class dependent

The SFR evaluation intentionally excludes the bus-fault rows because no scalar active-power disturbance is available without leaking the post-event response. This is not a missing-metadata issue; it is a model-applicability issue. Self-clearing bus faults remain useful for ML benchmarking, but they are outside the clean aggregate SFR disturbance class unless a follow-on trip or topology change is recorded with a defensible scalar disturbance proxy.

The expanded benchmark therefore uses load-step and generator-output-loss scenarios for SFR-evaluable comparisons. This creates a matched physics benchmark without retrofitting SFR to scenarios where its assumptions do not apply. In the expanded set, the SFR-compatible scenario classes contain 420 load-step scenarios and 400 generator-output-loss scenarios, with unsafe rates of 29.5% and 23.8%, respectively.

5.9. Expanded-benchmark SFR baselines expose model-class and information-set costs

SFR-class baselines apply the physical nadir threshold directly to predicted nadir, bypassing the learned-calibration step that produced the all-unsafe collapse in ML methods. This is one explicit operational advantage of physics-based decision rules in the rare-event regime.

Table 8 reports per-variant nadir MAE for both SFR-compatible scenario classes.

The headline pattern is consistent across classes: oracle accuracy is near 0.008–0.009 Hz, ambient DSE is about an order of magnitude worse, and 5 s event-triggered DSE matches oracle within about 0.001 Hz. These values separate three information sets: a full-response oracle upper bound, a pre-event ambient estimator, and an event-triggered estimator that commits after observing only the first part of the transient. SFR-oracle is used only as a model-class ceiling; the operational comparisons are the ambient and event-triggered variants.

The ambient-PSD finding remains based on dedicated stochastic-load calibration scenarios rather than a full-dataset PSD survey, but

Table 8
Expanded-benchmark SFR-family nadir MAE by scenario class. The table separates information sets and estimator variants: full-response oracle, pre-event ambient DSE, event-triggered DSE, noisy event-triggered DSE, and misspecified first-order event-window fits.

Variant	Load-step MAE (Hz)	Generator-output-loss MAE (Hz)
SFR-oracle	0.0076	0.0090
Ambient SFR-DSE	0.0798	0.0706
Event-triggered 1 s	0.0134	0.0165
Event-triggered 2 s	0.0131	0.0146
Event-triggered 5 s	0.0071	0.0077
Noisy event-triggered 1 s	0.0306	0.0133
Noisy event-triggered 2 s	0.0191	0.0177
Noisy event-triggered 5 s	0.0065	0.0064
Misspecified first-order 1 s	0.0206	0.0371
Misspecified first-order 2 s	0.0249	0.1376
Misspecified first-order 5 s	0.0044	0.0087

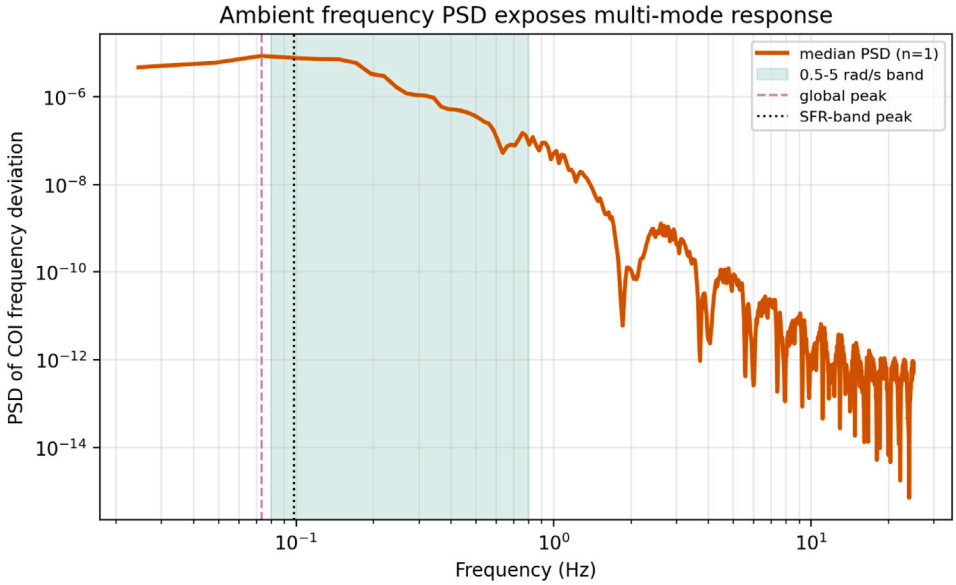


Fig. 4. Ambient center-of-inertia frequency-deviation PSD from the 300 s stochastic-load calibration scenario. The spectrum shows a persistent low-frequency peak and an SFR-band peak after detrending, suggesting that ambient SFR-DSE may face multi-mode model-class mismatch. The figure is used as an identifiability diagnostic rather than as a complete modal decomposition.

the expanded SFR batch confirms that the ambient estimator is weak at scale. Ambient SFR-DSE has nadir MAE 0.0754 overall, 0.0798 on load-step scenarios, and 0.0706 on generator-output-loss scenarios. This remains about an order of magnitude worse than 5 s event-triggered DSE, supporting the interpretation that pre-event ambient PSD fitting is limited by multi-mode response and effective-parameter identifiability. Fig. 4 shows the ambient COI frequency-deviation PSD from the calibration scenario, with a low-frequency peak and an SFR-band peak that motivate the multi-mode interpretation.

We extend the nameplate-anchor analysis from the five-scenario calibration subset to the full expanded SFR-compatible set. Across all SFR-oracle records, $M_{\mathrm{oracle}}/M_{\mathrm{nameplate,pretrip}}$ has mean 0.0090, median 0.0052, coefficient of variation 128.4%, and class-specific coefficients of variation 127.6% for load-step and 123.9% for generator-output-loss. This variability is far too large for a fixed per-system unit conversion and supports the conclusion that effective SFR inertia is contingency-dependent. Fig. 5 plots the per-scenario M-anchor ratios by scenario class and inertia regime.

This instability is physically plausible. Aggregate nameplate inertia sums stored kinetic energy across online machines, but an effective SFR inertia fitted to a particular COI response depends on modal participation: which machines swing coherently, which modes are excited by the disturbance location and type, and how governor and network dynamics shape the measured aggregate response. A single nameplate sum is therefore not expected to be a stable per-system conversion factor for every contingency.

These findings are not implementation errors. They identify operational baseline risks that must be controlled before using enhanced SFR-DSE as a reference method: model-class mismatch, multi-mode ambient response, unstable effective-inertia anchoring, and early-window identifiability.

5.10. Event-triggered SFR-DSE shows early-window identifiability costs

Event-triggered SFR-DSE avoids both ambient PSD amplitude dependence and nameplate-M anchoring by fitting effective parameters from the measured early disturbance response. Fig. 6 reports the full 1 s, 2 s, and 5 s window sweep by scenario class.

The headline pattern is monotone for the second-order event-triggered estimator: 2 s improves on 1 s, and 5 s improves further. The noisy event-triggered variants follow the same qualitative pattern: noise most strongly affects the 1 s window, while 2 s and 5 s fits are less sensitive because they contain more response information. The 5 s row reflects delayed nowcasting rather than early prediction: the nadir-forming response is largely visible by 5 s, so the fit commits after observing most of the post-disturbance dynamics.

A five-scenario calibration subset showed a non-monotone relationship, where the 2 s estimator was worse than the 1 s estimator, but

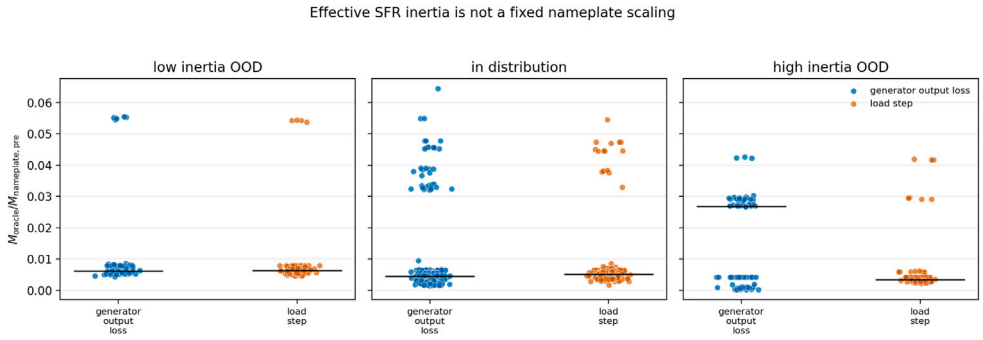


Fig. 5. Effective SFR inertia from the oracle fit relative to pre-event aggregate nameplate inertia on the expanded benchmark. Points show $M_{\text{oracle}}/M_{\text{nameplate,pretrip}}$ by scenario class and inertia regime. The broad spread indicates that nameplate aggregate inertia is not a stable per-system conversion factor for effective SFR inertia.

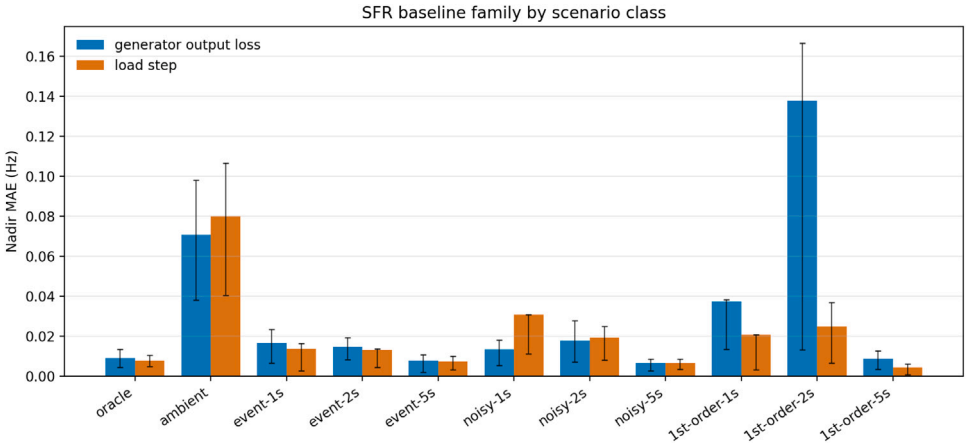


Fig. 6. Nadir MAE for the SFR baseline family on the expanded benchmark, stratified by scenario class. The variants separate information sets: full-response oracle, pre-event ambient DSE, event-triggered DSE, noisy event-triggered DSE, and misspecified first-order event-window fits.

that pattern does not persist at scale. The expanded-benchmark result is monotone in the expected direction: 2 s improves on 1 s, and 5 s improves further. The calibration-subset anomaly is therefore a small-sample artifact, and the expanded-benchmark monotone trend is the main result.

The misspecified first-order event-window model adds a parsimony test. Fig. 6 shows a sharp 2 s degradation for this first-order fit, even though the 5 s first-order envelope gives the best overall nadir MAE. The 2 s misspecified fit can lock onto a confidently wrong steady-state estimate when the window contains enough rising-edge information to constrain the fit but not enough settling behavior to identify the envelope. The 5 s parsimony advantage observed in the calibration subset persists in the overall mean and is sharpest on load-step scenarios. The interpretation is not that the first-order model is physically superior; it is that, once the nadir-forming trajectory is mostly observed and overshoot is modest, a lower-variance two-parameter envelope can outperform a more flexible three-parameter second-order fit for nadir estimation.

5.11. Event-triggered ML screening: A leak-free information set that avoids collapse

The all-unsafe collapse documented above is established under a regression/descriptor information set: each model predicts the post-event frequency trajectory from pre-event and event-descriptor inputs and derives a safety decision from the predicted nadir. A natural question is whether the collapse reflects a fundamental limit of artificial-intelligence-based DSA or the specific information set the model is given. We therefore add an event-triggered information set: the model observes only the first τ seconds of the measured center-of-inertia frequency response (a derived observation channel with an explicit observation mask), with all later samples masked and with no fault location, disturbance magnitude, generator identity, or inertia value provided. This information set is leak-free—the frequency nadir forms several seconds after the disturbance, so the first τ seconds do not include the final nadir or divergence outcome, and the model must infer severity from the early rate of change and slope. The measurement window is aligned with the event-triggered SFR-DSE baseline, while the SFR baseline also uses a characterized active-power disturbance; the comparison is therefore explicit about the shared response window and the remaining input differences.

Under this event-triggered information set the all-unsafe collapse largely disappears. Table 9 reports, for observation windows $\tau \in \{0.5, 1.0, 2.0\}$ s on all three systems, the number of the seven architectures that collapse to an all-unsafe decision at the −0.10 Hz threshold and the mean unsafe recall achievable within a 20% false-alarm budget. Where the descriptor-based information set yielded zero unsafe recall at every false-alarm budget, the event-triggered information set yields non-trivial operating points: on IEEE 39, two of seven models still collapse but the remainder reach 27%–39% unsafe recall at a 20% false-alarm budget; on NPCC and WECC no models collapse at $\tau \geq 1$ s and recall reaches 41% and 59% respectively. Recall increases monotonically with the observation window on all three systems – the expected early-warning-versus-delay tradeoff, since longer windows expose more of the nadir-forming response – mirroring the monotone improvement of the event-triggered SFR-DSE physics estimator. The $\tau = 2$ s window approaches delayed nowcasting in the same sense as the 5 s SFR-DSE fit, so $\tau \in \{0.5, 1.0\}$ s is the genuine early-warning regime.

Table 9
Event-triggered ML screening across systems and observation windows τ under a leak-free information set (only the first τ s of the measured center-of-inertia frequency response; no disturbance descriptors). "Collapse" is the number of the seven architectures producing an all-unsafe decision at the −0.10 Hz threshold on a majority of seeds; "Recall@20%FAR" is the mean unsafe recall across architectures within a 20% false-alarm budget. The descriptor-based information set yields zero recall at every budget, shown for reference.

System	Metric	τ=0.5 s	τ=1.0 s	τ=2.0 s
IEEE 39	Collapse @ −0.10 Hz (of 7)	2	2	2
IEEE 39	Recall @ 20% FAR	0.27	0.35	0.39
NPCC (140-bus)	Collapse @ −0.10 Hz (of 7)	0	0	0
NPCC (140-bus)	Recall @ 20% FAR	0.30	0.41	0.41
WECC (179-bus)	Collapse @ −0.10 Hz (of 7)	2	0	0
WECC (179-bus)	Recall @ 20% FAR	0.46	0.57	0.59

This result reframes the benchmark's message constructively. Artificial-intelligence-based DSA does not collapse because the task is infeasible; it collapses when the model is asked to decide safety from pre-event descriptors that do not pin down disturbance severity. Given an early measured response window aligned with the physics event-triggered estimator, common architectures recover usable, low-false-alarm operating points, and they do so consistently across a 39-bus, a 140-bus, and a 179-bus renewable-rich system. The constructive direction for artificial-intelligence-based DSA is therefore an event-triggered screening information set with explicit early-warning-versus-delay accounting, not a new architecture: the no-separable-winner finding also holds under the event-triggered information set, where at most five of twenty-one pairwise PR-AUC intervals exclude zero on any system and none do on NPCC.

We state two limits plainly. First, the collapse is reduced but not universally eliminated: at the lenient −0.05 Hz threshold, where the unsafe base rate is high, four of seven IEEE 39 models and all seven NPCC models still collapse, so the constructive claim is threshold-dependent. Second, the recovered operating points are usable rather than strong – IEEE 39 reaches 27%–39% unsafe recall at a 20% false-alarm budget depending on the window – so event-triggered ML screening is best treated as a complementary prioritization layer, not a replacement for physics-based assessment.

5.12. Runtime and operational considerations

The ML models are fast at inference once trained. On the expanded benchmark, GPU batched per-sample inference time ranges from about 0.04 ms for LSTM and Transformer+LSTM to 0.56 ms for LiquidMamba on the evaluated workstation, excluding simulator and data-loading time. The SFR event-triggered variants require little computation after the measurement window is available; their limiting factor is not numerical fit time but the observation delay. A 5 s event-triggered fit is near oracle quality in this benchmark, but it is a delayed nowcast rather than a preventive early-warning decision and may be too late for protection actions that operate on sub-second timescales.

The operational recommendation is therefore conservative. For operators with reliable disturbance characterization and a validated aggregate response model, SFR-class methods with explicit information-set documentation remain the appropriate reference baseline for frequency-security screening. ML-based DSA is better treated as a complementary screening or prioritization layer until it demonstrates stable low-false-alarm unsafe recall under multi-seed, safety-aware evaluation. Scaling to larger systems would require revalidating SFR-compatibility constraints, disturbance definitions, and per-system stress envelopes before transferring any benchmark conclusions. Deployment would also require bad-data handling, PMU/SCADA time-alignment checks, missing-measurement logic, and validation on the target system rather than direct transfer of the IEEE 39-bus numerical rankings.

6. Conclusion

DSA-SafetyBench is a benchmark and methodology contribution for artificial-intelligence-based dynamic security assessment. It supplies a task definition, scoring protocol, per-scenario record format, repeated-seed protocol, and reference baseline suite for testing whether machine-learning models produce operationally meaningful security decisions. Under that benchmark, regression-derived machine-learning safety decisions can be degenerate even when trajectory metrics look acceptable. Explicit three-class safety supervision improves unsafe-event ranking, but five-seed evaluation does not establish a statistically separable architecture winner. On the expanded benchmark, denser unsafe coverage improves unsafe-event ranking but still leads to all-unsafe hard decisions under recall-weighted F-score, target-recall, and fixed-alarm-budget analyses.

The simulator audit shows that the reported synthetic history fields are not causally wired into the transient simulator. Future history-aware DSA work therefore needs either repaired simulator pathways or measured operational histories where the causal connection is physically present.

The descriptor-based collapse is robust along two natural axes. The full seven-model five-seed run on the expanded benchmark removes seed-42 preselection bias; no architecture separates statistically from the others. Threshold sensitivity at −0.05 Hz, −0.10 Hz, and −0.20 Hz shows the all-unsafe collapse persists across the operationally realistic threshold range and is not an artifact of the chosen label, and calibration-policy robustness across F2-opt, target recall 0.90 and 0.95, and false-alarm budgets 5%–20% shows no monotone post-hoc calibration recovers a useful operating point from the descriptor-based scores.

The collapse is, however, information-set-dependent rather than fundamental to artificial-intelligence-based dynamic security assessment. Under a leak-free event-triggered information set – each model sees only the first 0.5–2 s of the measured center-of-inertia frequency response, aligned with the event-triggered physics estimator's measurement window – the all-unsafe collapse is largely removed and usable low-false-alarm operating points emerge across the IEEE 39, NPCC, and WECC systems, subject to threshold- and system-dependent limits, with recall increasing monotonically as the observation window lengthens. The benchmark's constructive message is therefore that artificial-intelligence-based dynamic security assessment collapses when asked to decide safety from pre-event descriptors that do not pin down disturbance severity, and that an event-triggered screening information set with explicit early-warning-versus-delay accounting is the path on which common architectures become operationally useful.

The physics baselines add the other side of the same message. Swing-frequency-response-class methods have explicit scenario-class and information-set limits, and enhanced dynamic state estimation is not trivial in practice: ambient estimation remains weak, nameplate inertia is an unstable effective-parameter anchor, and event-triggered estimation approaches oracle quality only when the operator can wait long enough to observe most of the nadir-forming response. Together, the base and expanded benchmark results show that

machine-learning dynamic-security claims require safety-aware metrics, multi-seed uncertainty estimates, threshold reporting, and physics-baseline information-set accounting before operational superiority over physics methods can be defended.

7. Limitations

The descriptor-based reference benchmark, with its full ambient SFR-DSE physics family, is on the IEEE 39-bus system. NPCC (140-bus) and WECC (179-bus, renewable-rich) enter as external systems for the leak-free event-triggered comparison, not as full descriptor-based or ambient-SFR replications. Their streams are deliberately scoped: a single governor-agnostic disturbance class each – generator-trip for NPCC and load-step at convergence-screened buses for WECC, because the generator-output-loss generator requires a TGOV1N governor model that neither system provides – together with short ambient windows and no ambient SFR-DSE family. The cross-system event-triggered result should therefore be read as protocol-level external validation under a common early-response information set rather than as a complete multi-system benchmark. NPCC's unsafe events are additionally concentrated in the largest generator trips, which limits unsafe-class diversity on that system. Extending any system to a full descriptor-based and ambient-SFR benchmark would require a calibrated stress envelope, split manifest, and SFR-compatibility audit of the kind already completed for IEEE 39.

The constructive event-triggered result is threshold-dependent. At the lenient -0.05 Hz threshold the all-unsafe collapse partially returns (four of seven IEEE 39 models and all seven NPCC models), and the recovered IEEE 39 unsafe recall is modest (27%–39% at a 20% false-alarm budget, depending on the observation window). Event-triggered ML screening is therefore a complementary prioritization layer that benefits from a longer observation window at the cost of decision delay, not a standalone replacement for physics-based assessment.

The base benchmark study uses five random seeds. That is sufficient for the paper's negative claim that the available data do not support a statistically separable architecture winner. Five seeds may not be sufficient for a strong positive claim in datasets where architectures separate cleanly; additional seeds are warranted in those settings. The paired bootstrap also treats scenarios as exchangeable within each seed after stratified splitting. Scenarios that share a disturbance class, inertia regime, or source bus/generator are not fully independent physical experiments. Class-stratified reporting partly mitigates this issue, but a block bootstrap by disturbance class or operating regime would be more conservative in larger follow-on studies.

The SFR-DSE conclusions are based on the expanded SFR-compatible IEEE 39-bus stream under load-step and generator-output-loss disturbances. They do not generalize to bus faults, protection-driven topology changes, or other systems without repeating the same disturbance-compatibility and DSE-identifiability checks. The ambient PSD interpretation also remains an interpretive check rather than a full modal identification. The observed low-frequency and SFR-band peaks suggest multi-mode response, but confirming their modal origin would require small-signal eigenanalysis and participation-factor analysis of the specific ANDES case. The present paper uses the PSD result to motivate SFR-DSE caution, not to claim a complete modal decomposition.

Finally, all scenarios are generated in the open-source ANDES dynamic-simulation environment. Operational systems may differ because of protection coordination, governor and AGC implementation details, measurement quality, topology changes, and operator procedures. The contribution is therefore a reproducible benchmark and evaluation framework, not a claim that the exact numerical rankings will transfer unchanged to deployed control rooms.

8. Reproducibility

The expanded benchmark uses ANDES 2.0.0 with the built-in IEEE 39-bus case name IEEE 39. The ML split is 60/20/20 train/validation/test, and the five training seeds are 7, 42, 99, 123, and 2025. The expanded benchmark contains 820 scenarios, with 420 load-step and 400 generator-output-loss cases; the test split contains 164 scenarios, including 44 unsafe cases.

The evaluated workstation uses an AMD Ryzen 7 8845HS CPU with 8 cores and 16 logical processors, 42 GB system memory, and an NVIDIA GeForce RTX 4070 Laptop GPU. Per-sample ML inference times on the expanded test set are approximately 0.04 ms for LSTM and Transformer+LSTM, 0.31 ms for LiquidPure, and 0.56 ms for Liquid-Mamba. The artifact package includes per-scenario ML records for both the top-four base-benchmark run and the full seven-model expanded-benchmark run, SFR-family records, calibrated thresholds, the paired scenario bootstrap script, threshold-sensitivity and calibration-policy comparison scripts that operate on the saved per-scenario records without retraining, figure scripts, scenario manifests, a prediction-record schema, a benchmark scoring script, and a case-inventory file documenting which ANDES systems are reported, screened, or future extensions.

For reference, the true nadir score on the base-benchmark seed-42 test split achieves PR-AUC 0.722 and F2-opt 0.833, confirming that the safety labels are recoverable from an accurate nadir signal even though learned regression-derived scores lose the unsafe-event ranking.

CRediT authorship contribution statement

Mohamed Shamseldein: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author used OpenAI Codex/ChatGPT to assist with language editing, LaTeX formatting, code review, and drafting of analysis scripts. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Funding statement

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.egyai.2026.100811.

Data availability

Data will be made available on request.

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